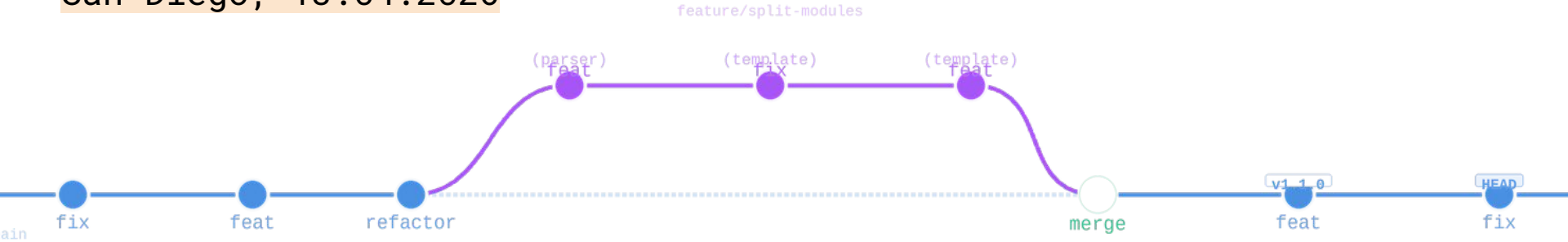
Conventional Commits

Meaningful versioning and automated changelong

Software Citation Workshop
San Diego, 13.04.2026



Lucas de Sá
Humboldt Research Fellow
✉ lucas.desa@uni-heidelberg.de

v1.0.0 doi:...

v1.1.0 doi:...

v1.1.1 doi:...

v1.1.2 doi:...

v1.2.0 doi:...

v1.3.0 doi:...

v2.0.0 doi:...

?

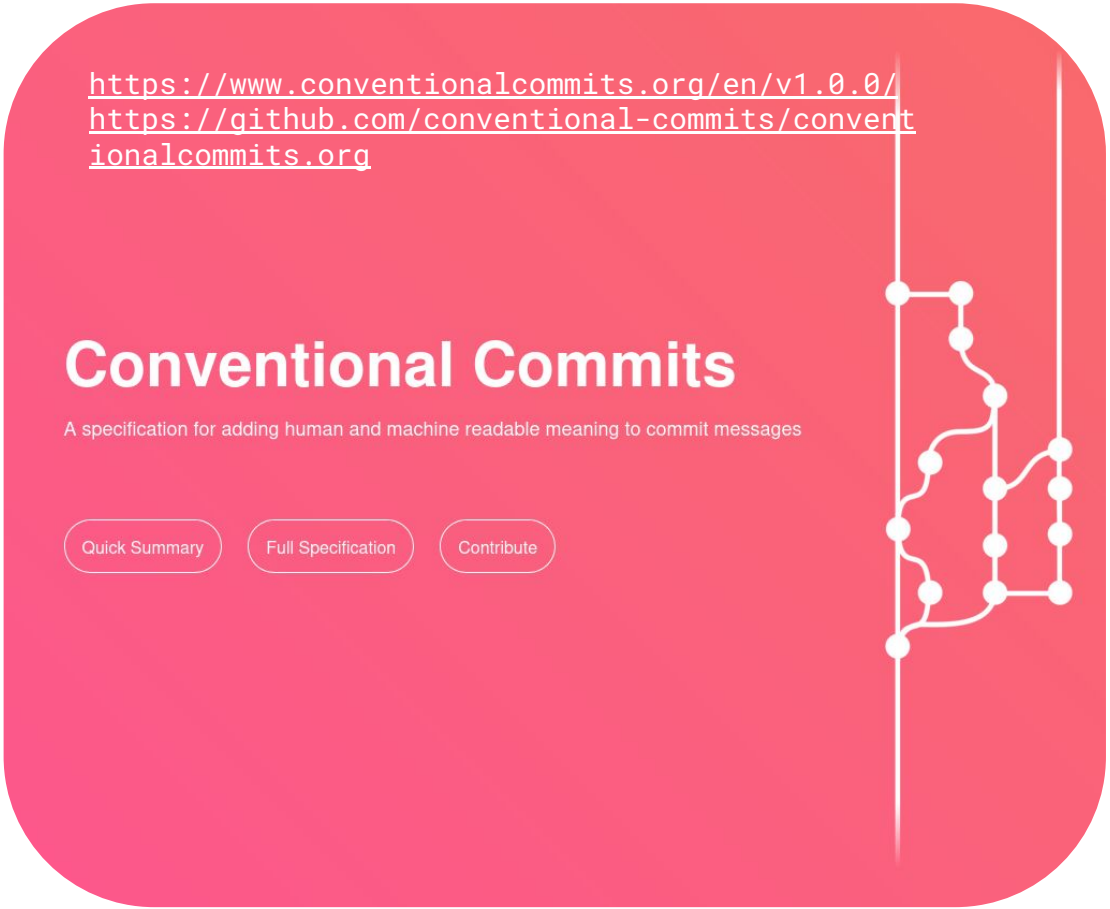
Versioning

citability ≠ readability

if a paper is published with v1.1.1, can I use v1.1.2? What about v1.2.0? v2.0.0?

where should I look to know if v1.1.1 had a bug? or if it had missing physics?

https://www.conventionalcommits.org/en/v1.0.0/
https://github.com/conventional-commits/conventionalcommits.org



Conventional Commits (CC)

offers a standard for informative histories and automatic release+changelog generation

built on **atomic commits**

CC relies on a simple structure and only 3 types

```
<type>[scope]: <description>

[body]

[footer]
```

types

fix: patches a bug

feat: introduces a new feature

feat!: introduces a breaking/major change

[footer] BREAKING CHANGE: same as feat!

types + scope

-> **Semantic Versioning** (SemVer)

extra types for git log

types (@commitlint/config-conventional)

build: build system, dependencies…

ci: continuous integration (eg, GitHub actions)

docs:documentation

perf: performance improvement

refactor: restructuring code, same behavior

revert: reverse previous change

style: formatting, whitespace, linting…

test: adding or fixing tests

chore: other maintenance

+fix, feat, feat!

https://github.com/conventional-changelog/commitlint/tree/master/%40commitlint/config-conventional

SemVer

https://github.com/semver/semver

vMAJOR.MINOR.PATCH

- PATCH: fix: backwards-compatible bug fix
- MINOR: feat: backwards-compatible new feature
- MAJOR: feat!: breaking change

release-please

https://github.com/semver/semver

automatic PR with changelong based on CC+SemVer, commit-linking, scope-separation.

conventional commits

github.com/LucasMDeSa/conv_commits_ucsd2026

commit → rebase → push → action fires → release PR → merge → GitHub release

git rebase -i HEAD~N

```
d8c0364 feat!: restructure author fields to support multiple auth
pick b95efe4 chore: add release-please GitHub Action
pick f1e100f chore: add README with workshop mention
pick 7aa194d chore(main): release 0.1.0
pick 71e6b93 chore: add README with workshop mention

# Rebase f043d9c..71e6b93 onto f043d9c (5 commands)
#
# Commands:
# p, pick <commit> = use commit
# r, reword <commit> = use commit, but edit the commit message
# e, edit <commit> = use commit, but stop for amending
# s, squash <commit> = use commit, but meld into previous commit
# f, fixup [-C | -c] <commit> = like "squash" but keep only the previous
#                   commit's log message, unless -C is used, in which case
#                   keep only this commit's message; -c is same as -C but
#                   opens the editor
# x, exec <command> = run command (the rest of the line) using shell
# b, break = stop here (continue rebase later with 'git rebase --continue')
# d, drop <commit> = remove commit
# l, label <label> = label current HEAD with a name
# t, reset <label> = reset HEAD to a label
# m, merge [-C <commit> | -c <commit>] <label> [# <oneline>]
#         create a merge commit using the original merge commit's
#         message (or the oneline, if no original merge commit was
#         specified); use -c <commit> to reword the commit message
# u, update-ref <ref> = track a placeholder for the <ref> to be updated
#                       to this position in the new commits. The <ref> is
#                       updated at the end of the rebase
#
# These lines can be re-ordered; they are executed from top to bottom.

If you remove a line here THAT COMMIT WILL BE LOST.

ever, if you remove everything, the rebase will be aborted.
```

LucasMDeSa/conv_commits_ucsd2026 on Apr 10

chore(main): release 0.1.0 #1

Merged

🤖 I have created a release *beep boop* 0.1.0 (2026-04-11) ⚠ BREAKING CHANGES 'author_l...

`main` ← `release-please--b...`

autorelease: tagged

v0.1.0 Latest

0.1.0 (2026-04-11)

⚠ BREAKING CHANGES

- 'author_last' and 'author_first' keys are removed. Callers must migrate to authors=[{"last": ..., "first": ...}].

Features

- add format_mla formatter for MLA 9th edition (aac8319)
- add volume, number, and pages fields to BibTeX output (7438a85)
- restructure author fields to support multiple authors (d8c0364)

Bug Fixes

- strip whitespace from BibTeX citation keys (a196614)

Assets 2

Source code (zip)	19 hours ago
Source code (tar.gz)	19 hours ago